LLVM & Industry

What is a compiler

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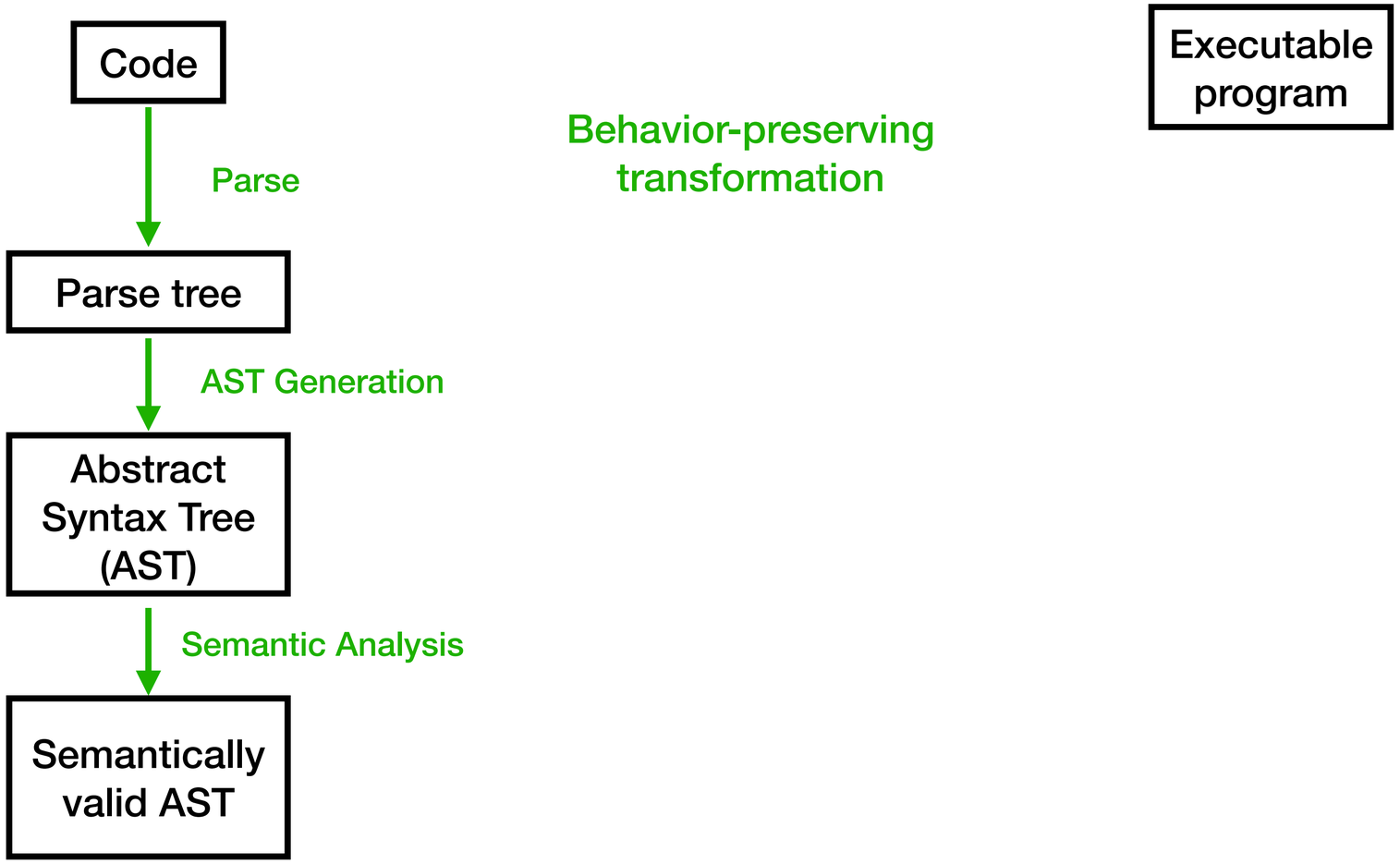

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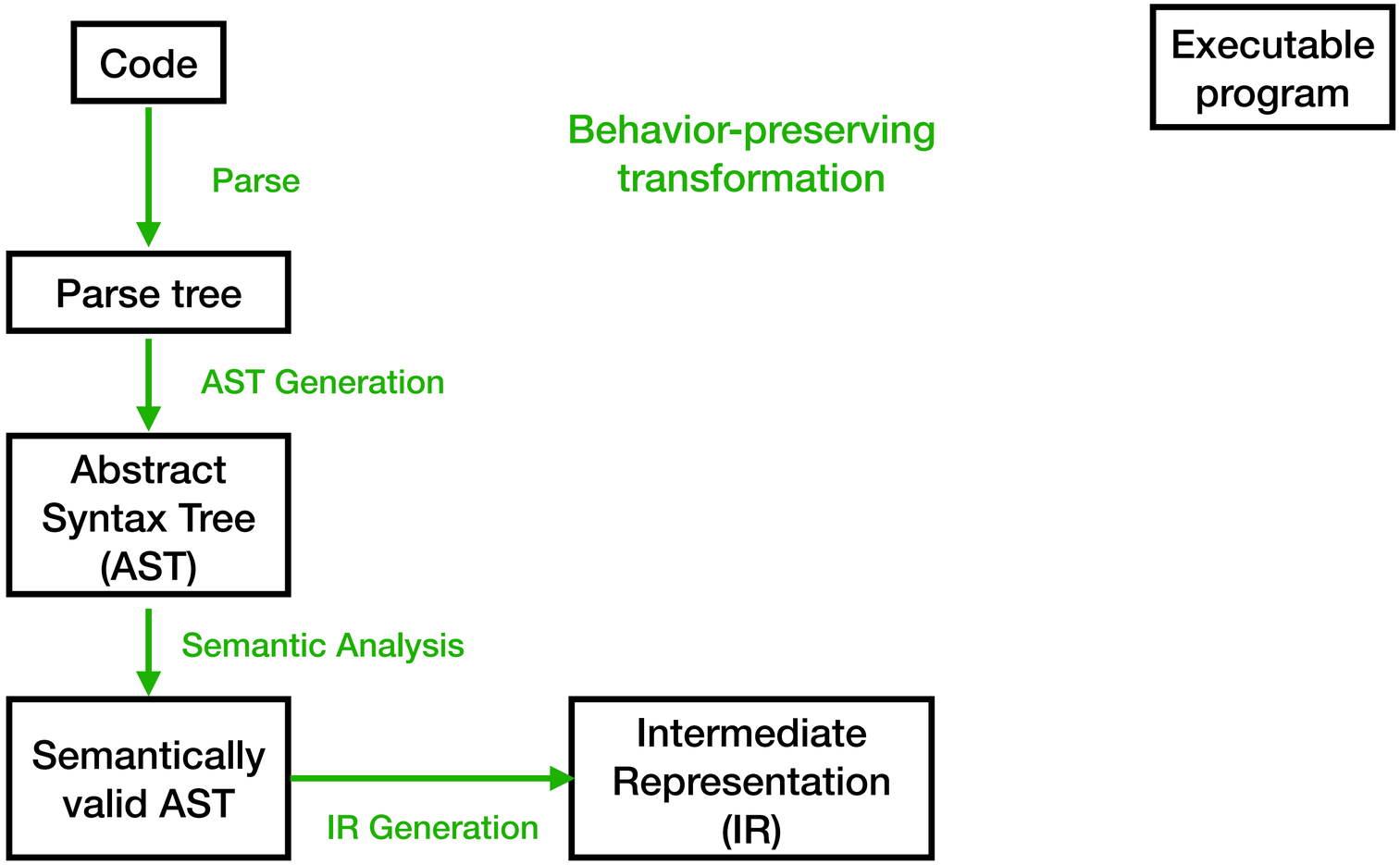

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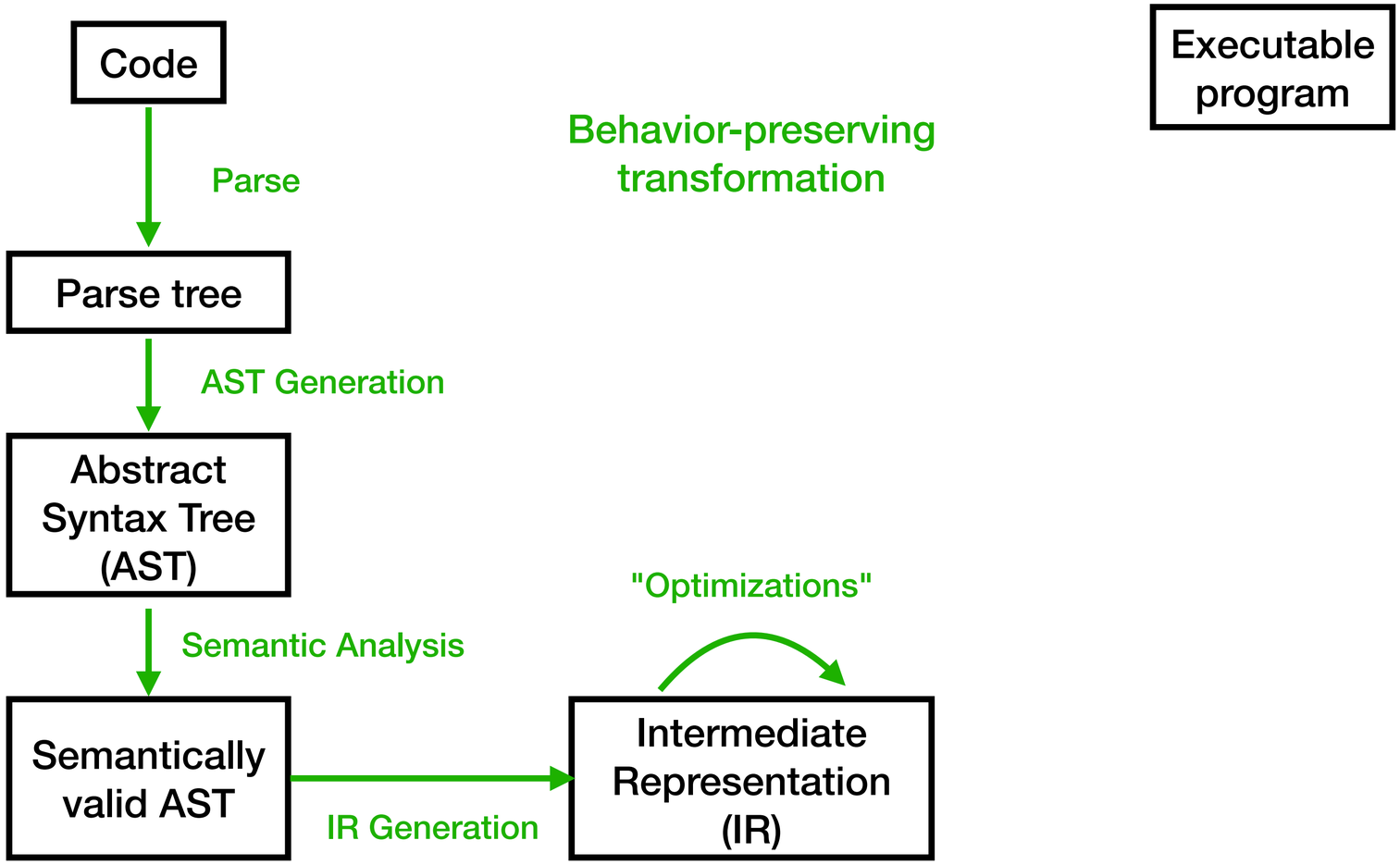

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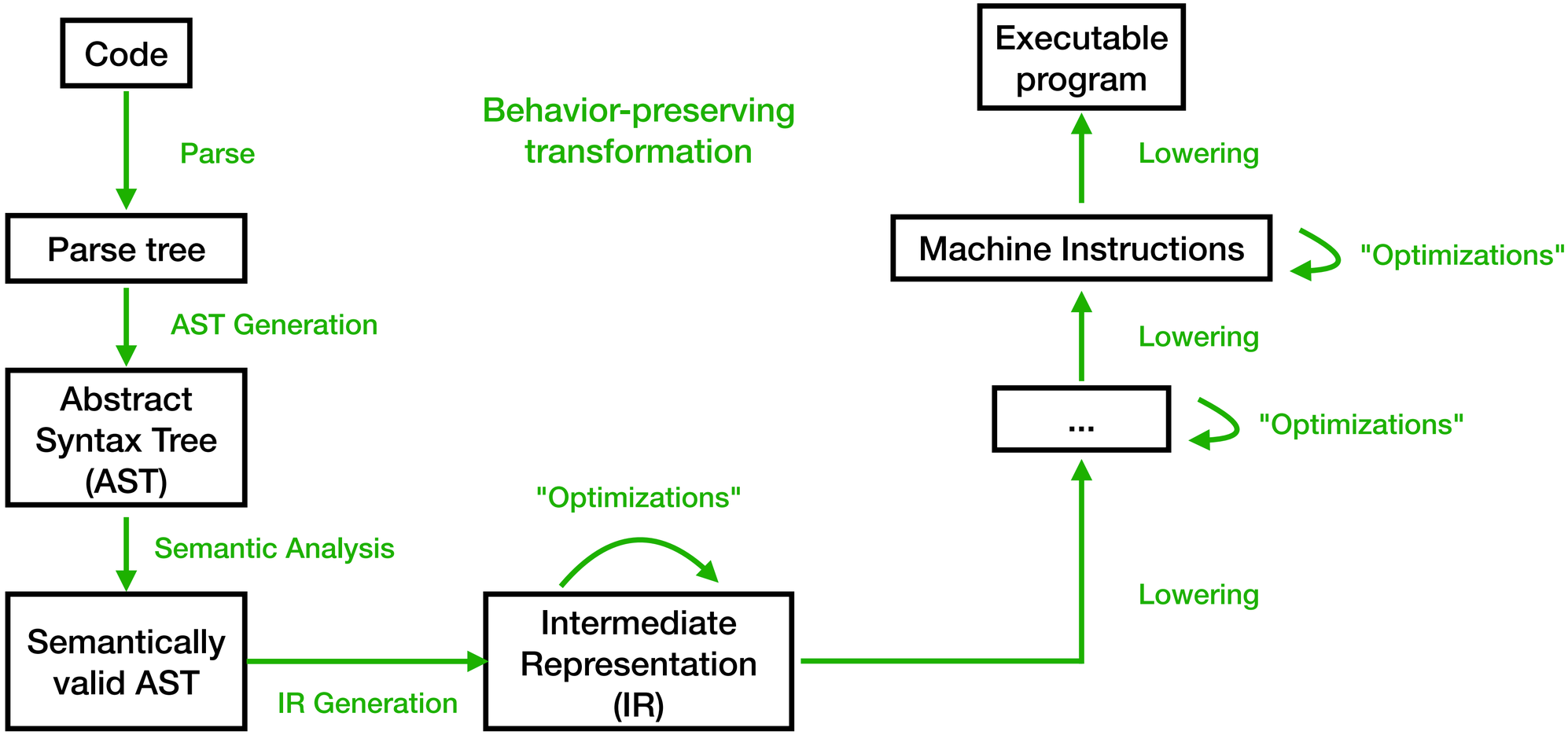

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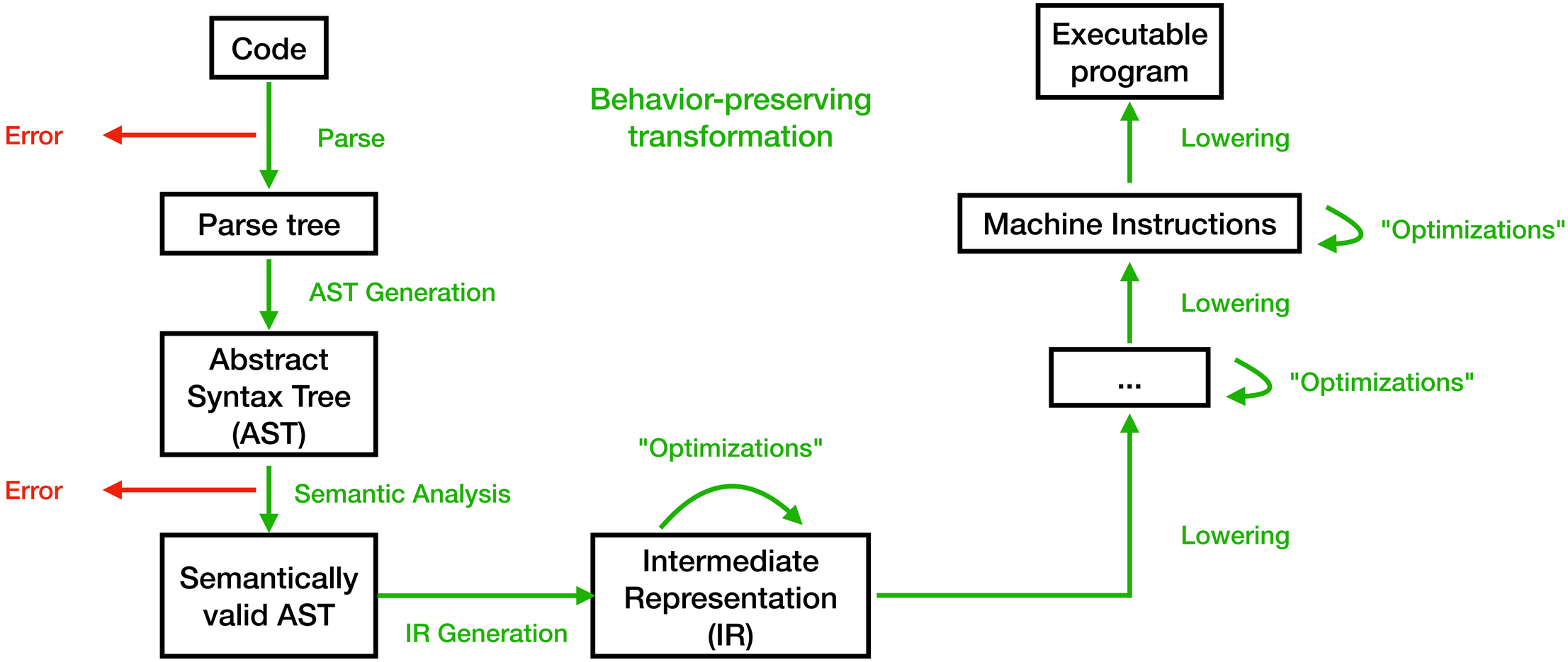

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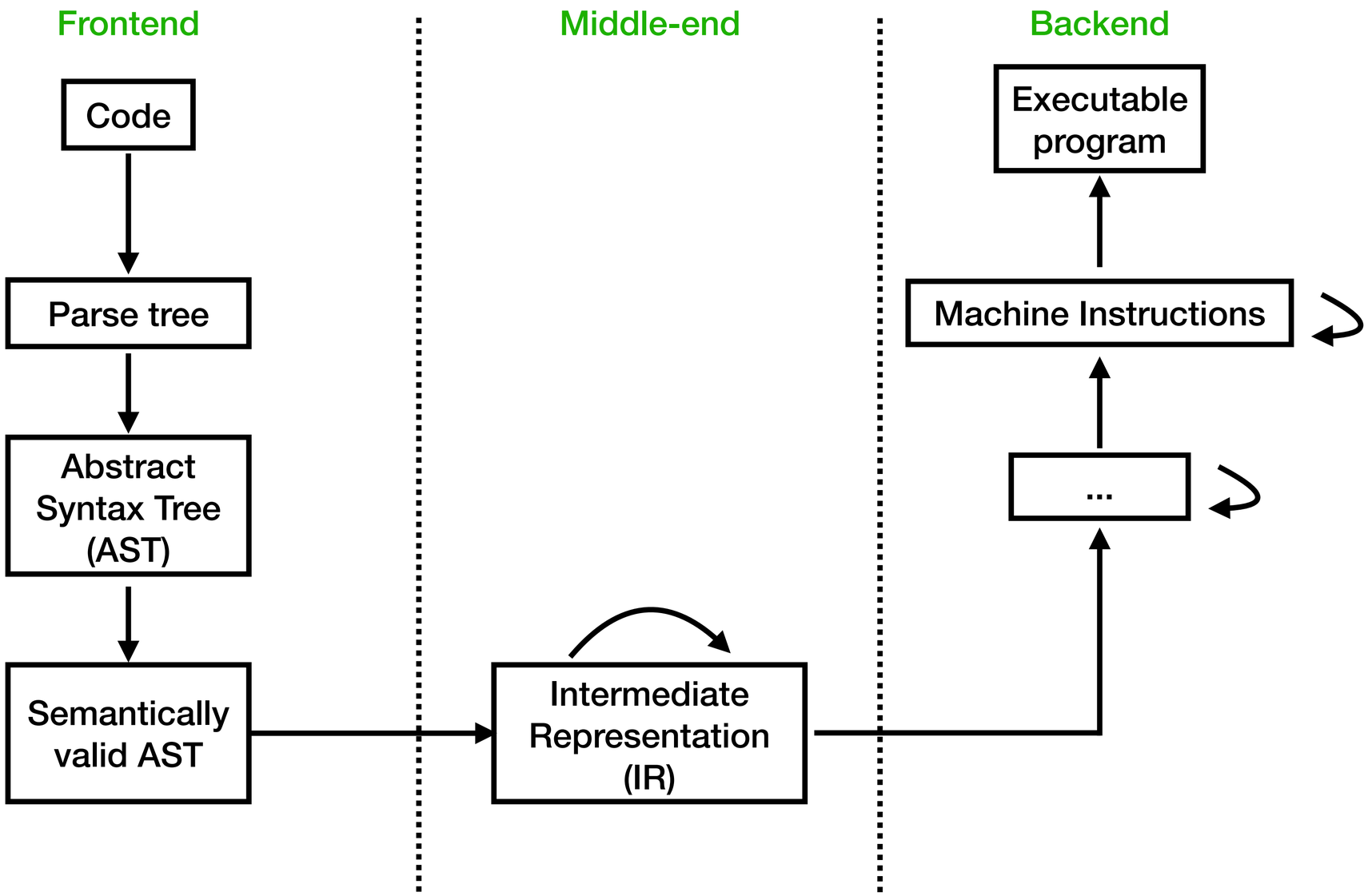
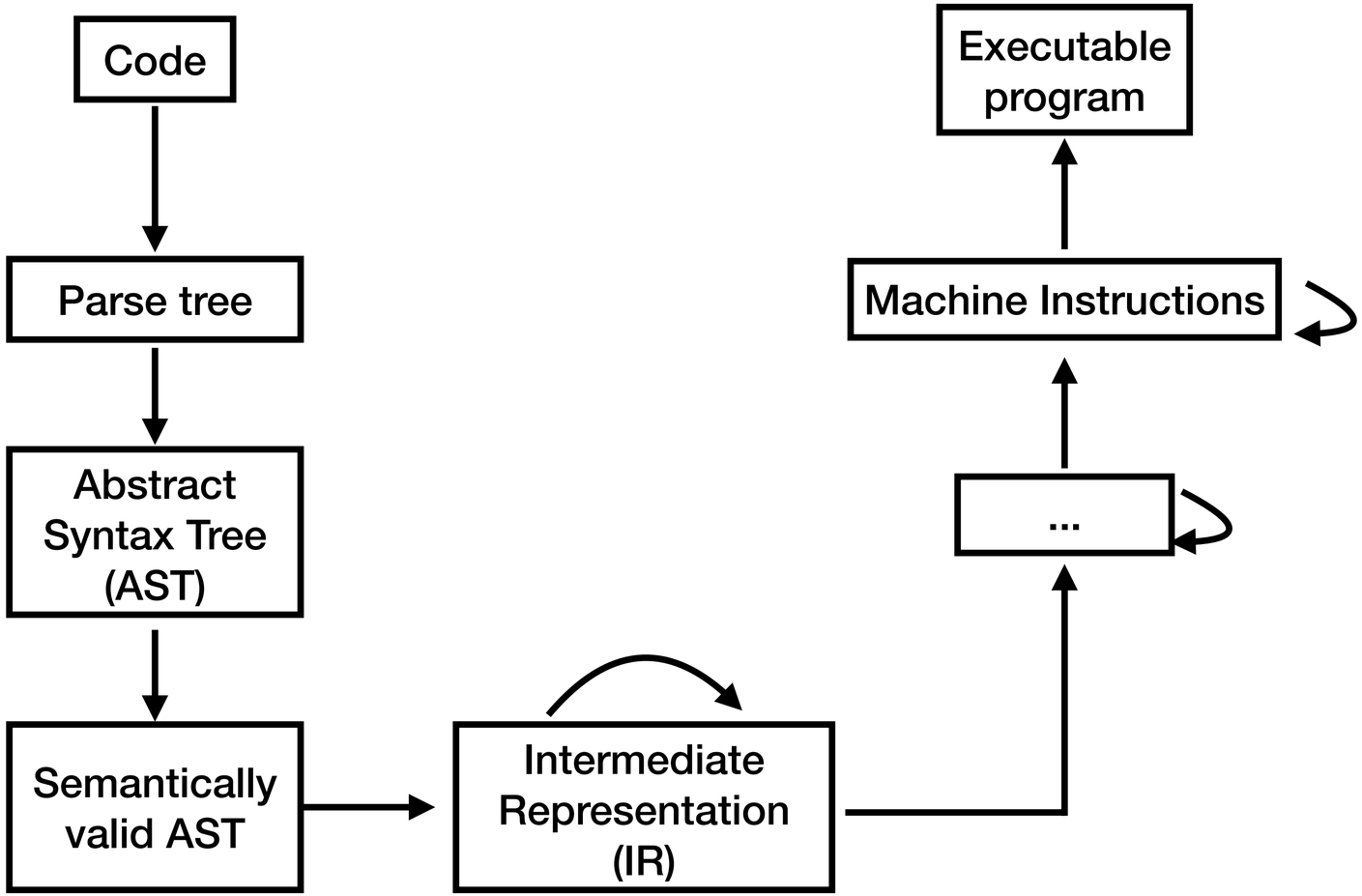


LLVM

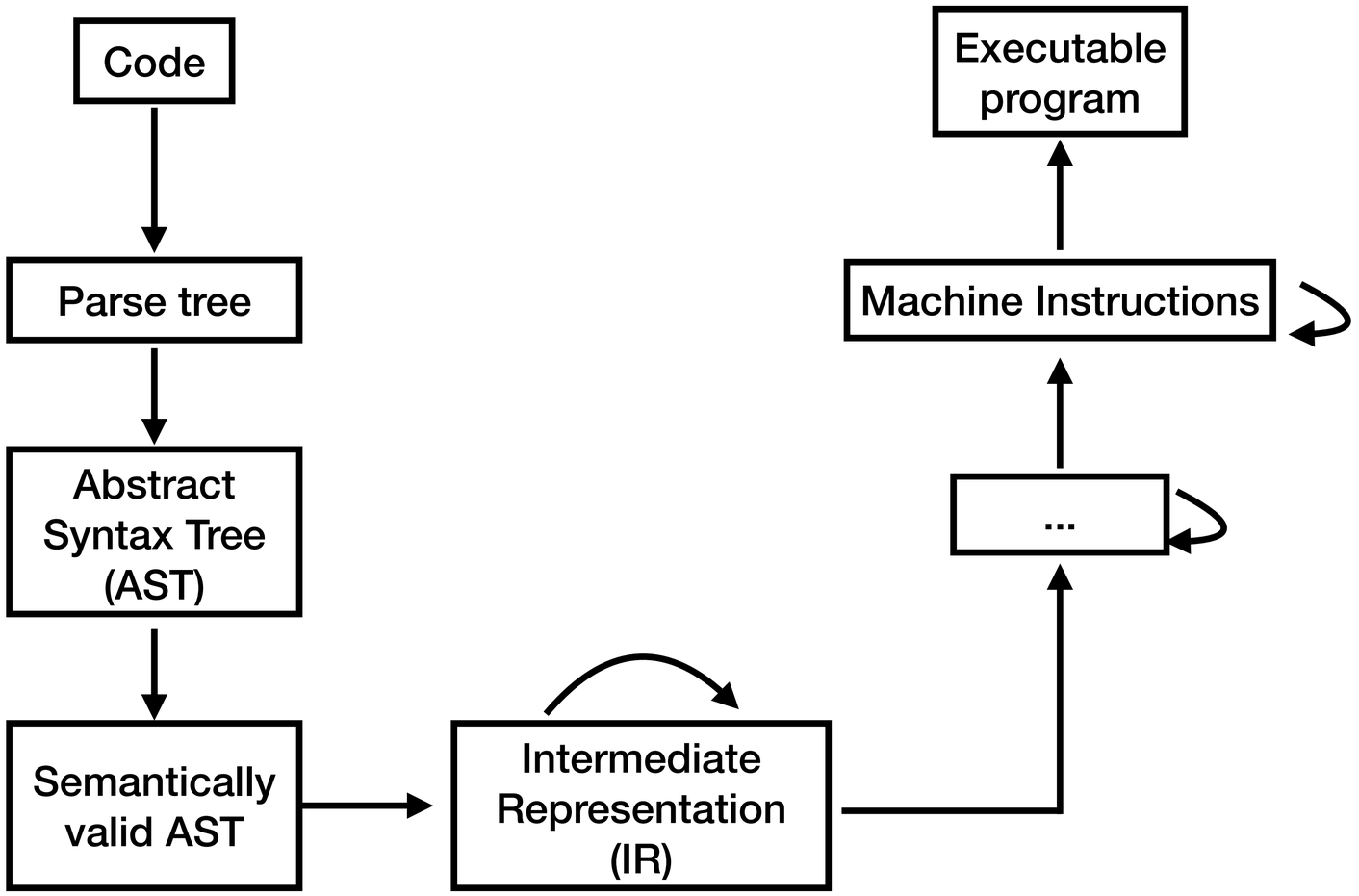

Code
Parse tree
Abstract Syntax Tree (AST)
Semantically valid AST
Intermediate Representation (IR)
...
Machine Instructions
Executable program

LLVM

Open source compiler project

Created in 2000 by Chris Lattner.

Permissive license.



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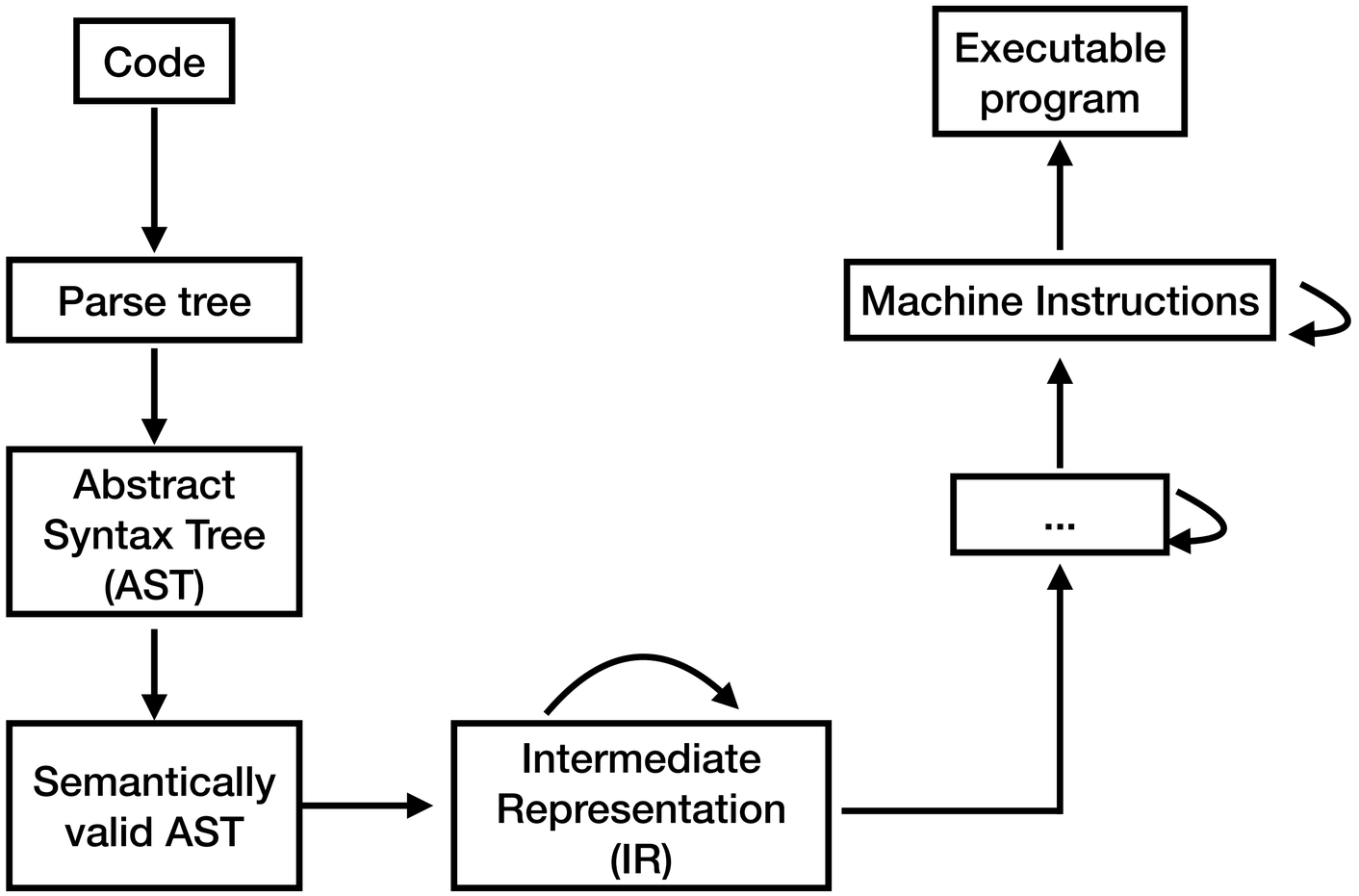

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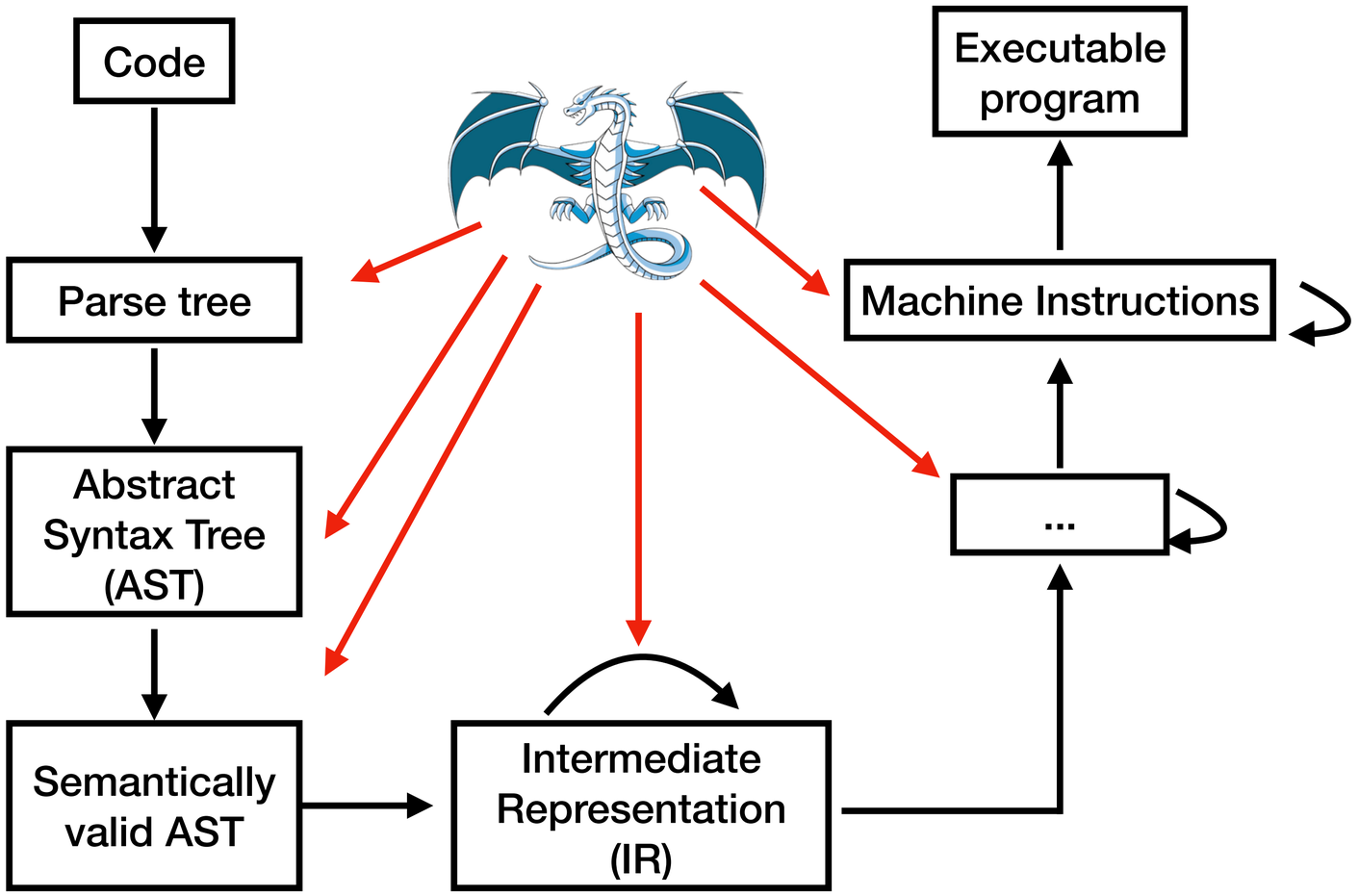
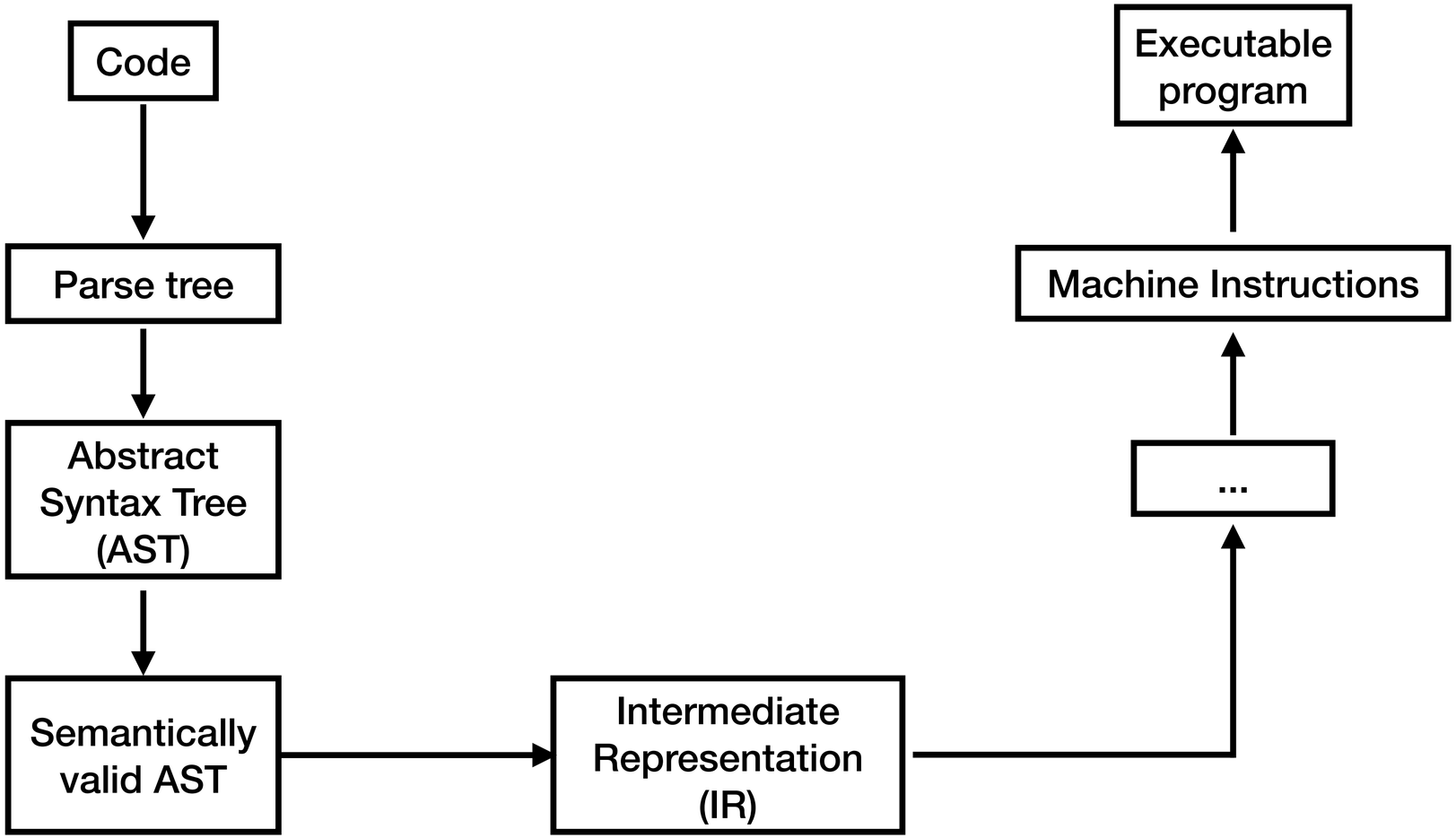


Supporting a new language



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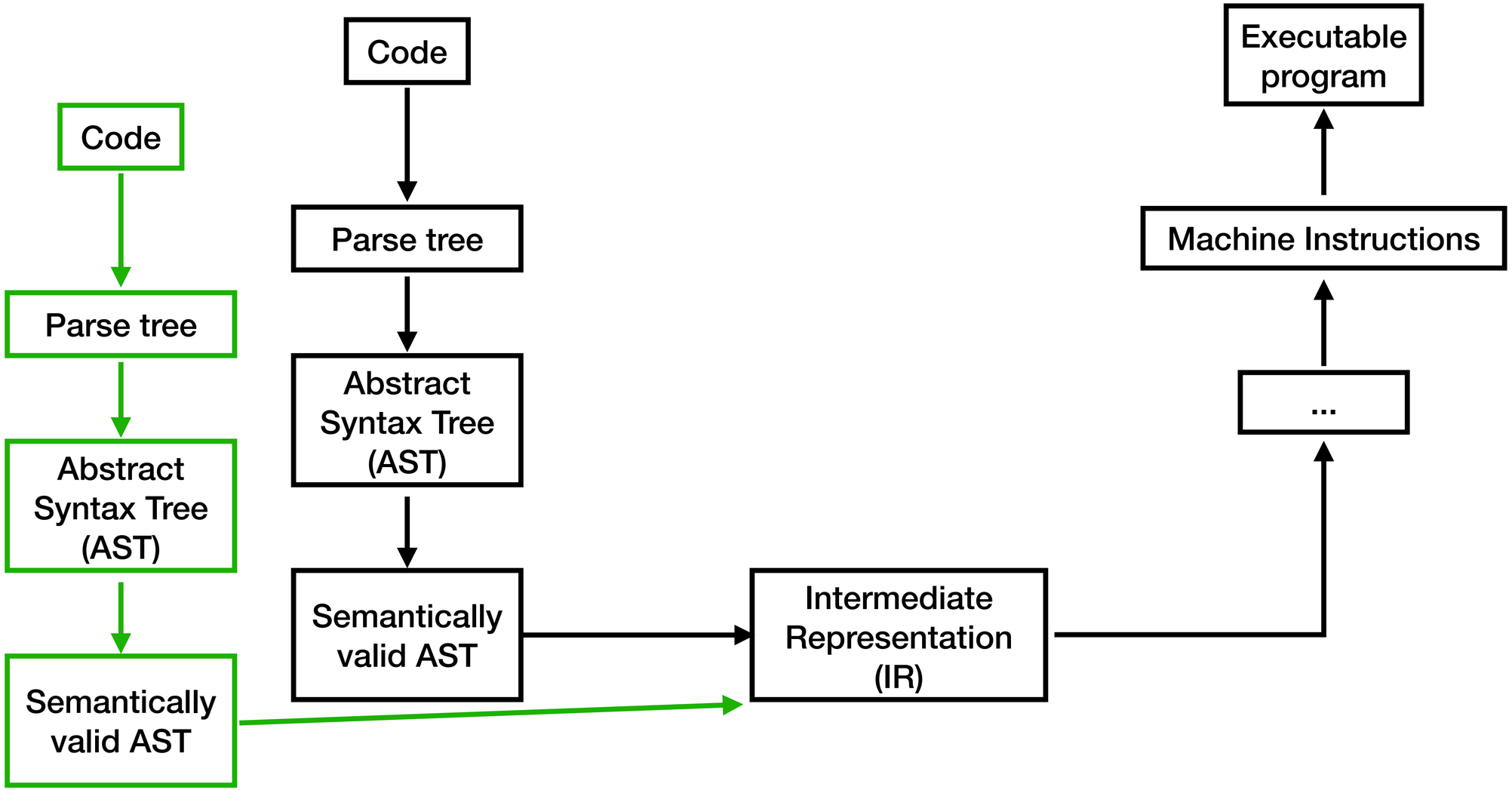
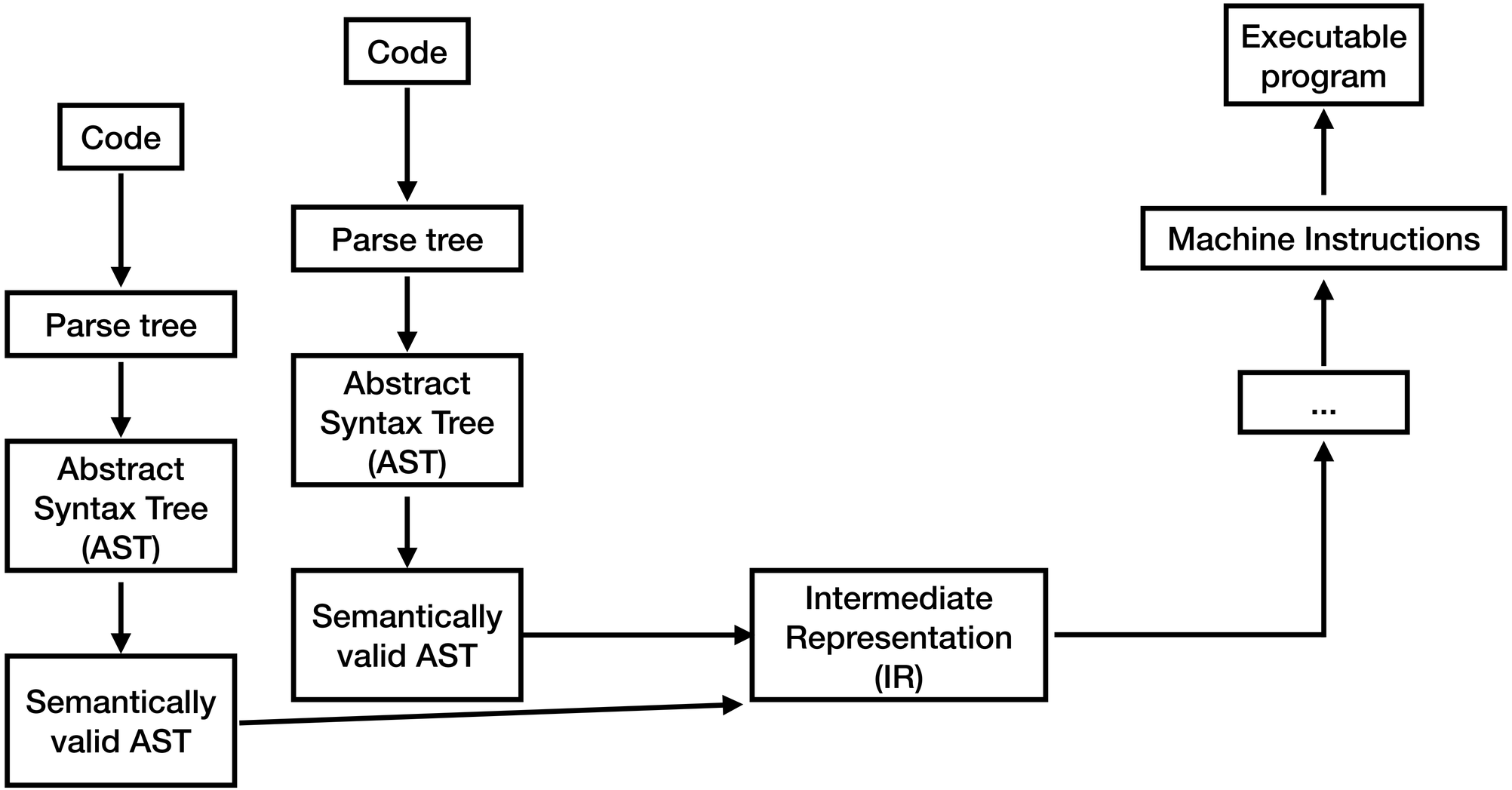


Supporting a new target



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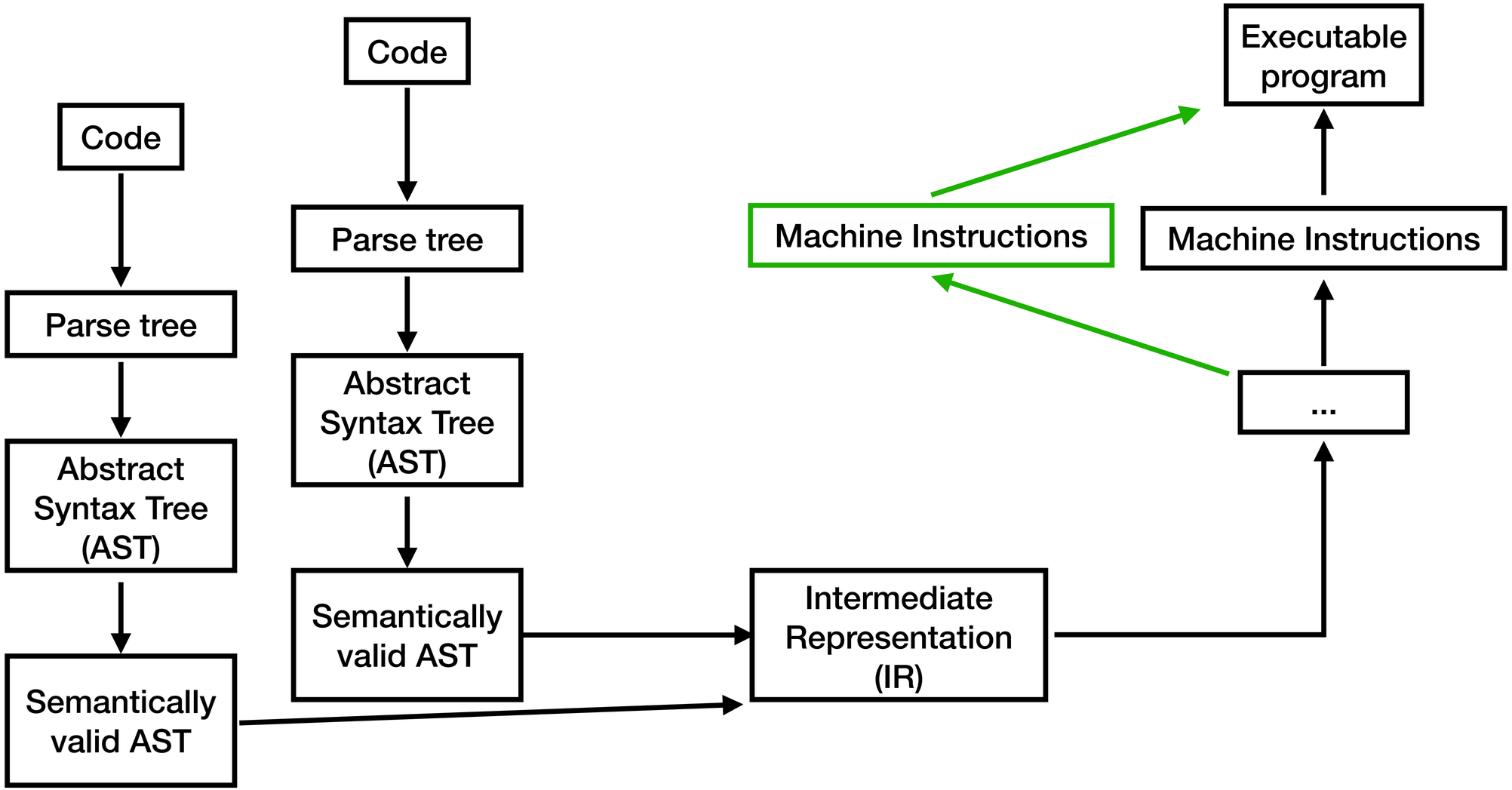

The perfect storm

LLVM

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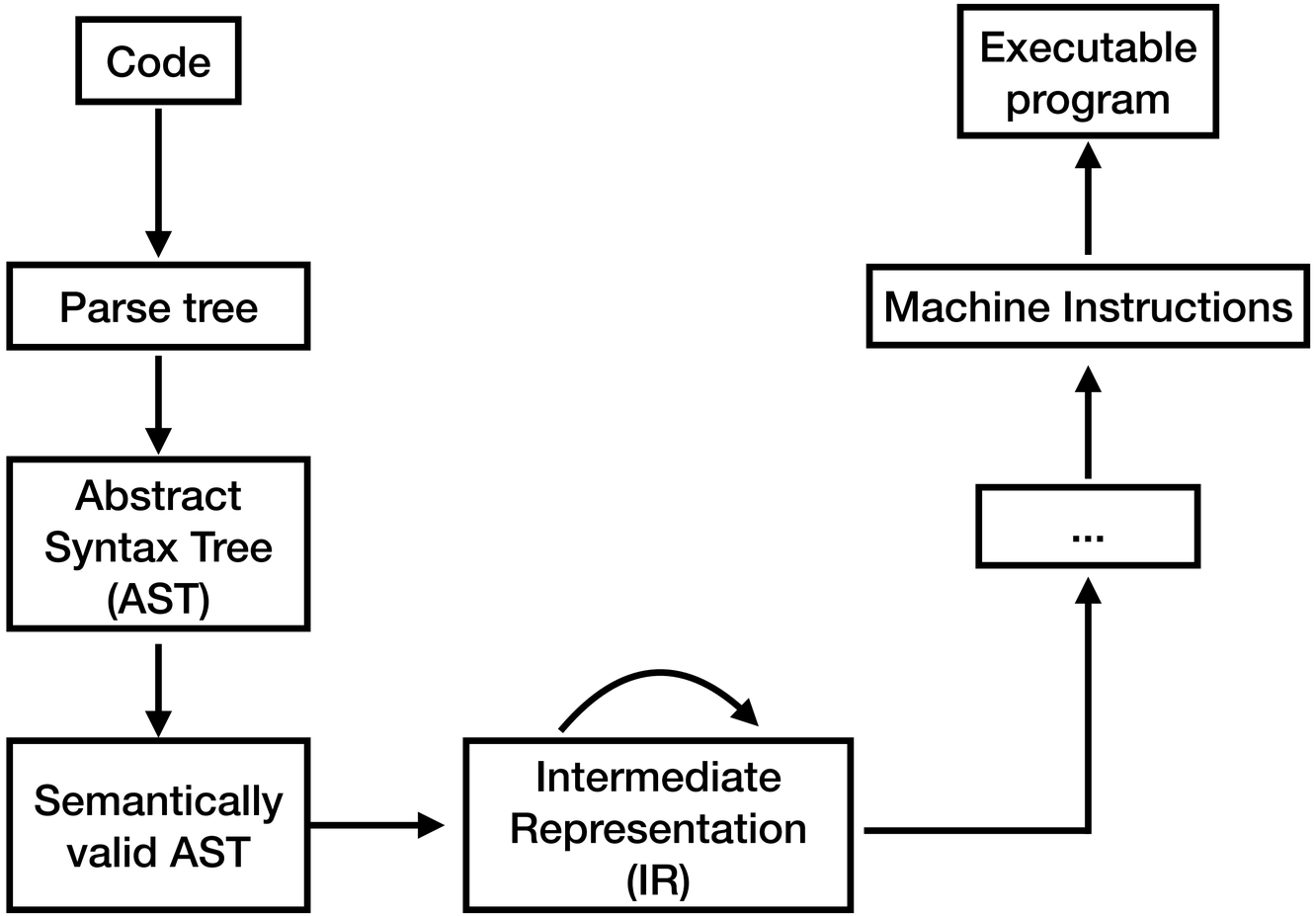
Industry-academia relationships.

LLVM & industry

CUDA -> C++ extensions

GPU backend

IR optimizations



LLVM & industry

Google

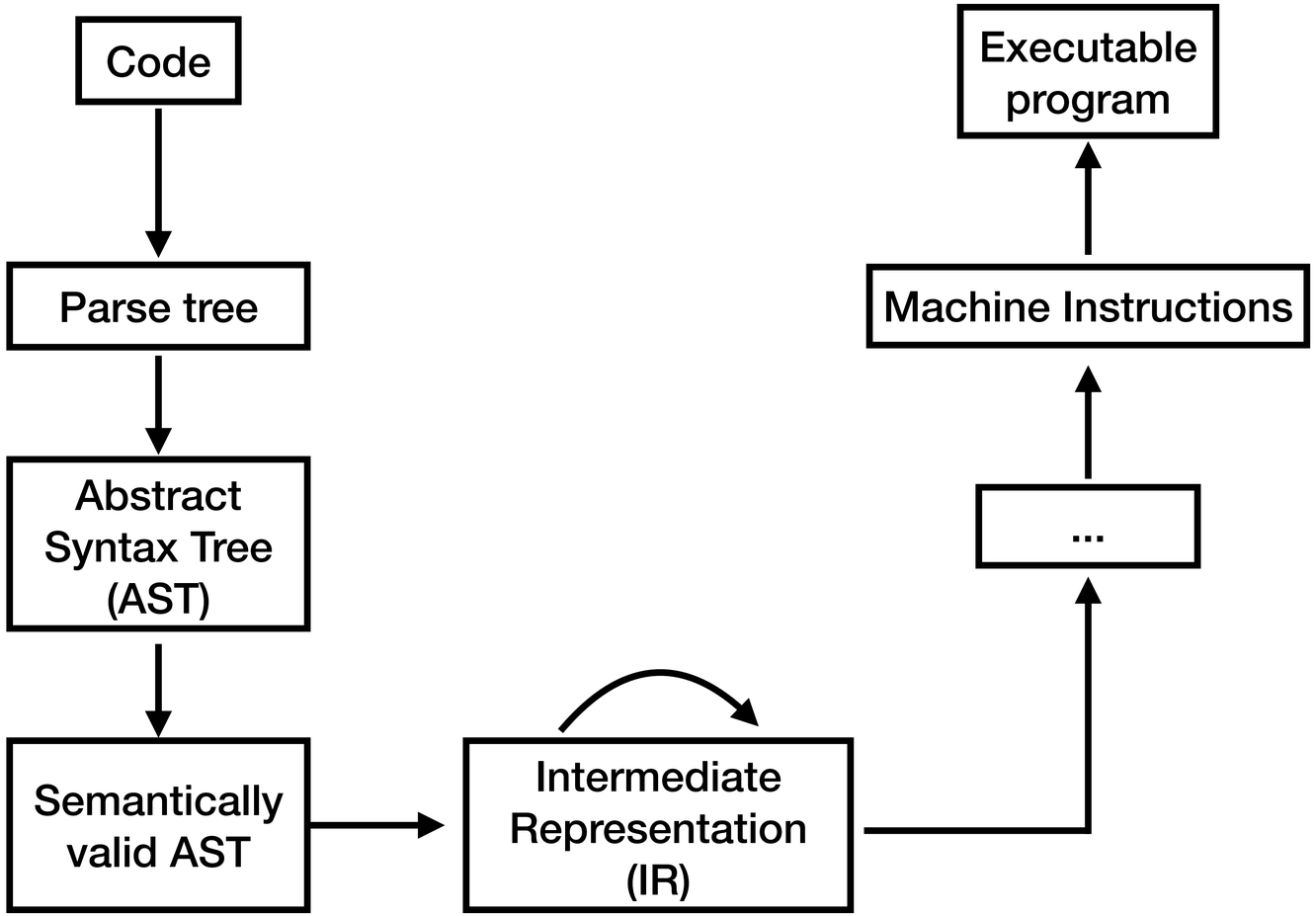
C++ / Carbon / Go frontends

IR optimizations

Libc / Libcxx

LLDB (Debuggers)

TPU backends

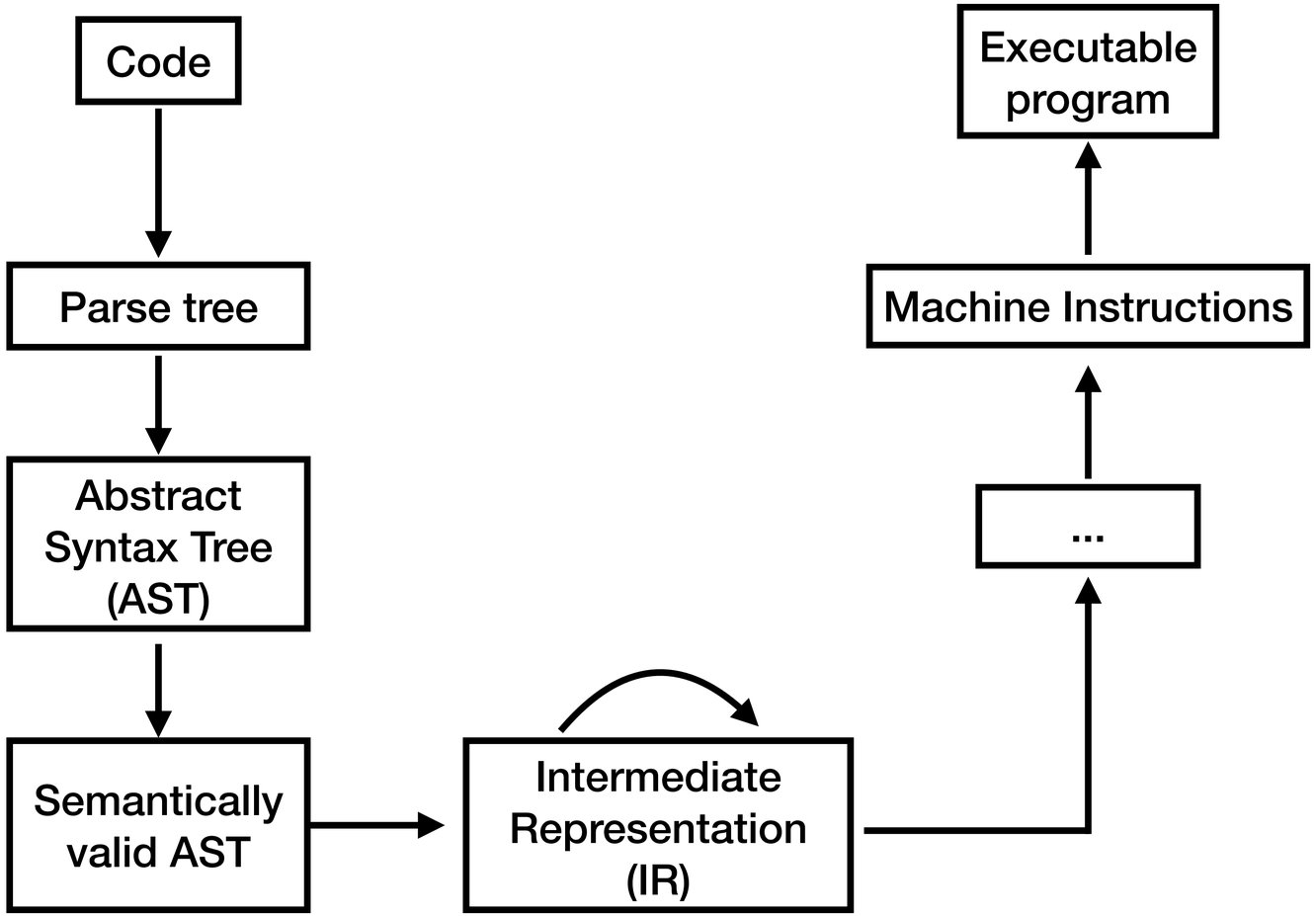


LLVM & industry

Nintendo PlayStation

Debuggers

Toolchains



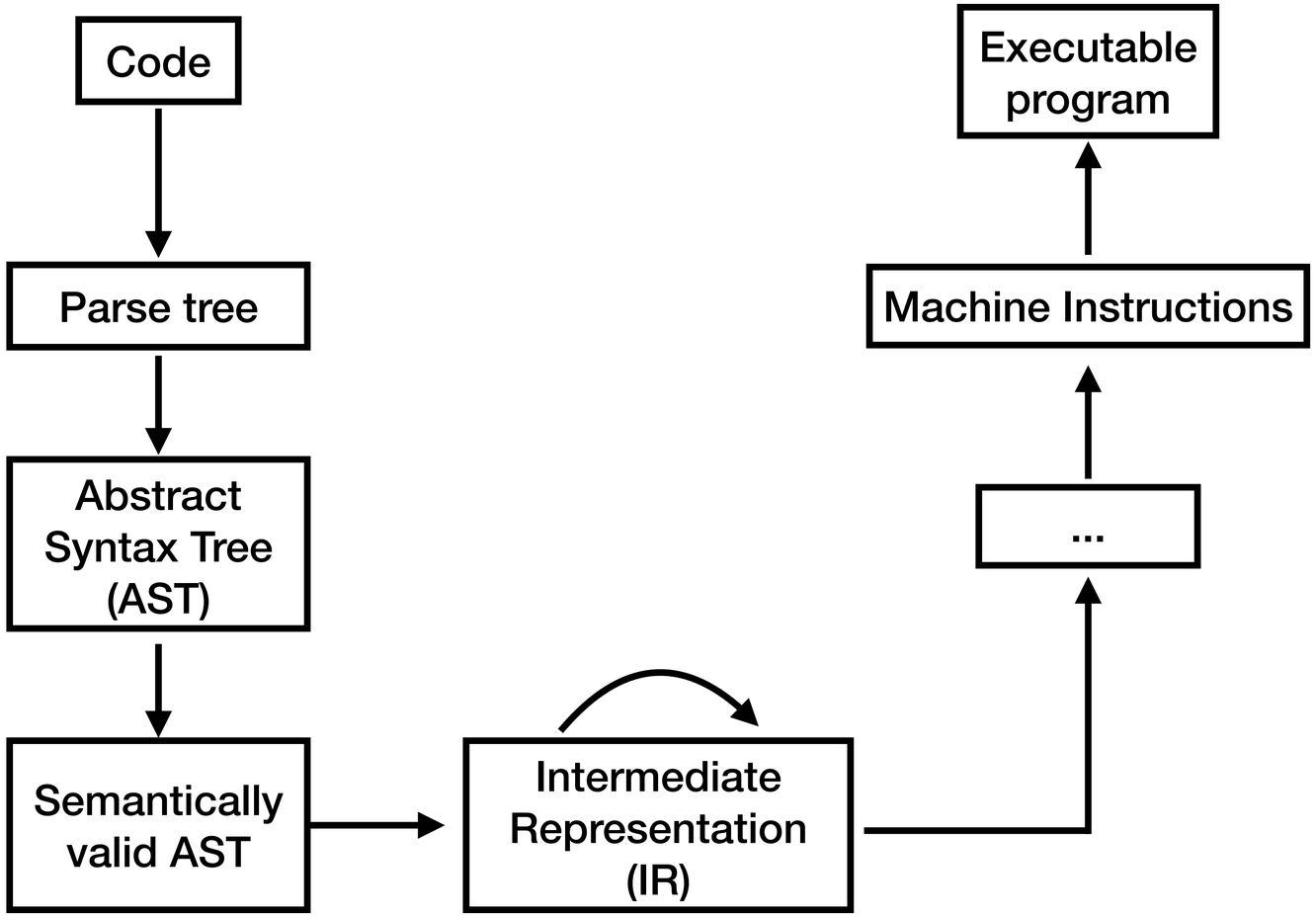
LLVM & industry

Meta

Debuggers

Swift support

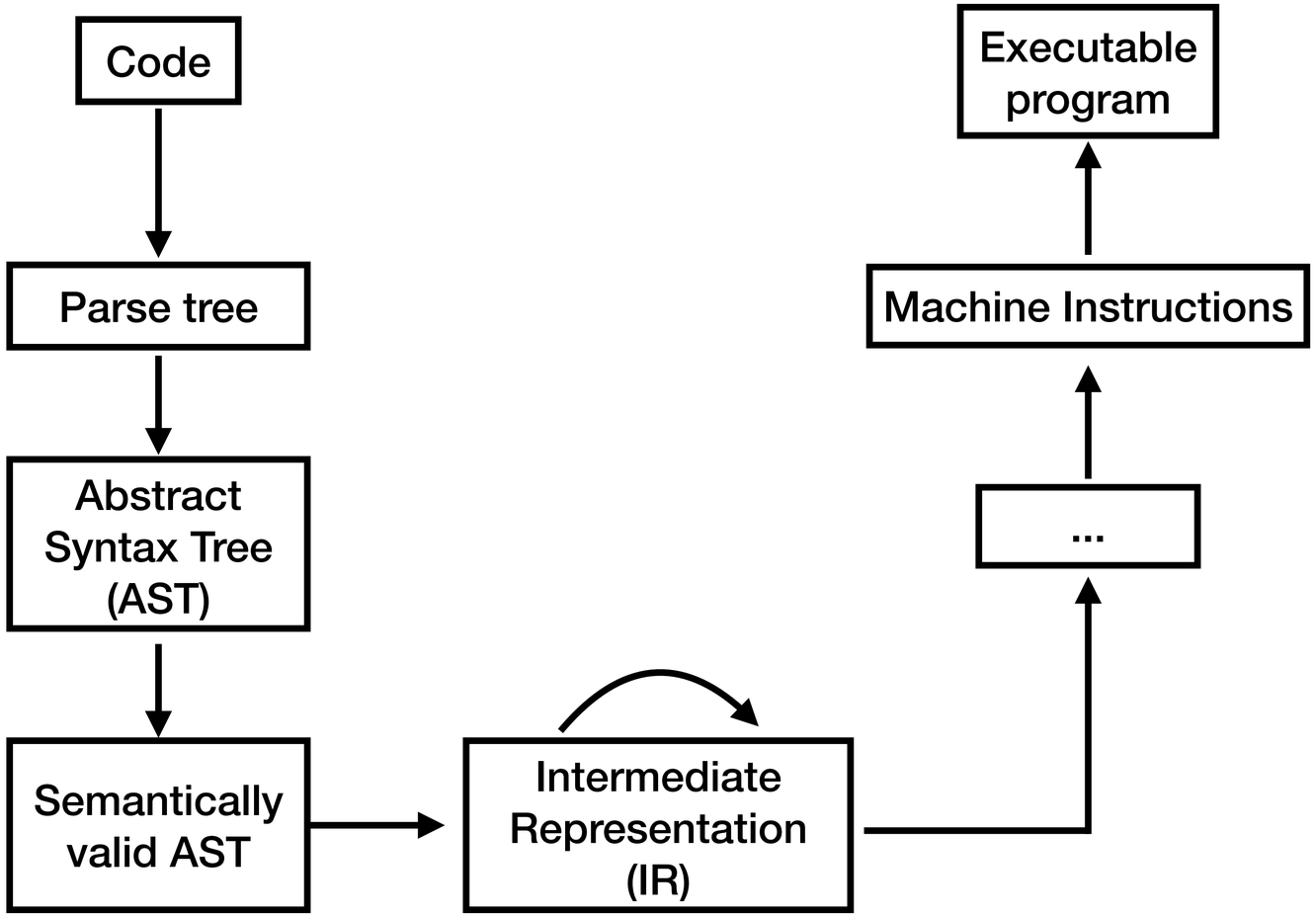
Clang improvements



LLVM & industry

Backends for their new processors

IR optimizations

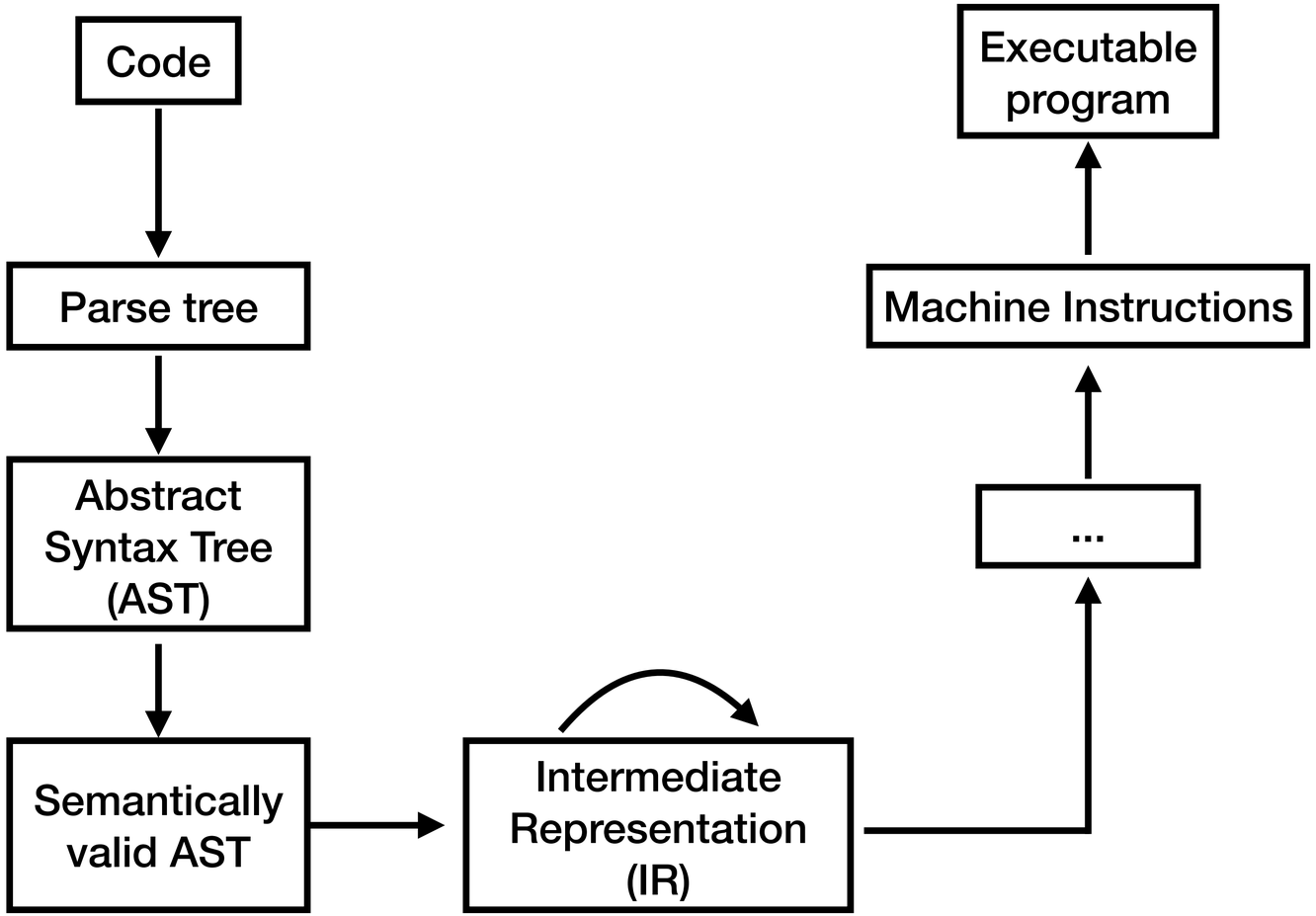


LLVM & industry

XILINX altera

C -> VHDL compilers

IR + Backend



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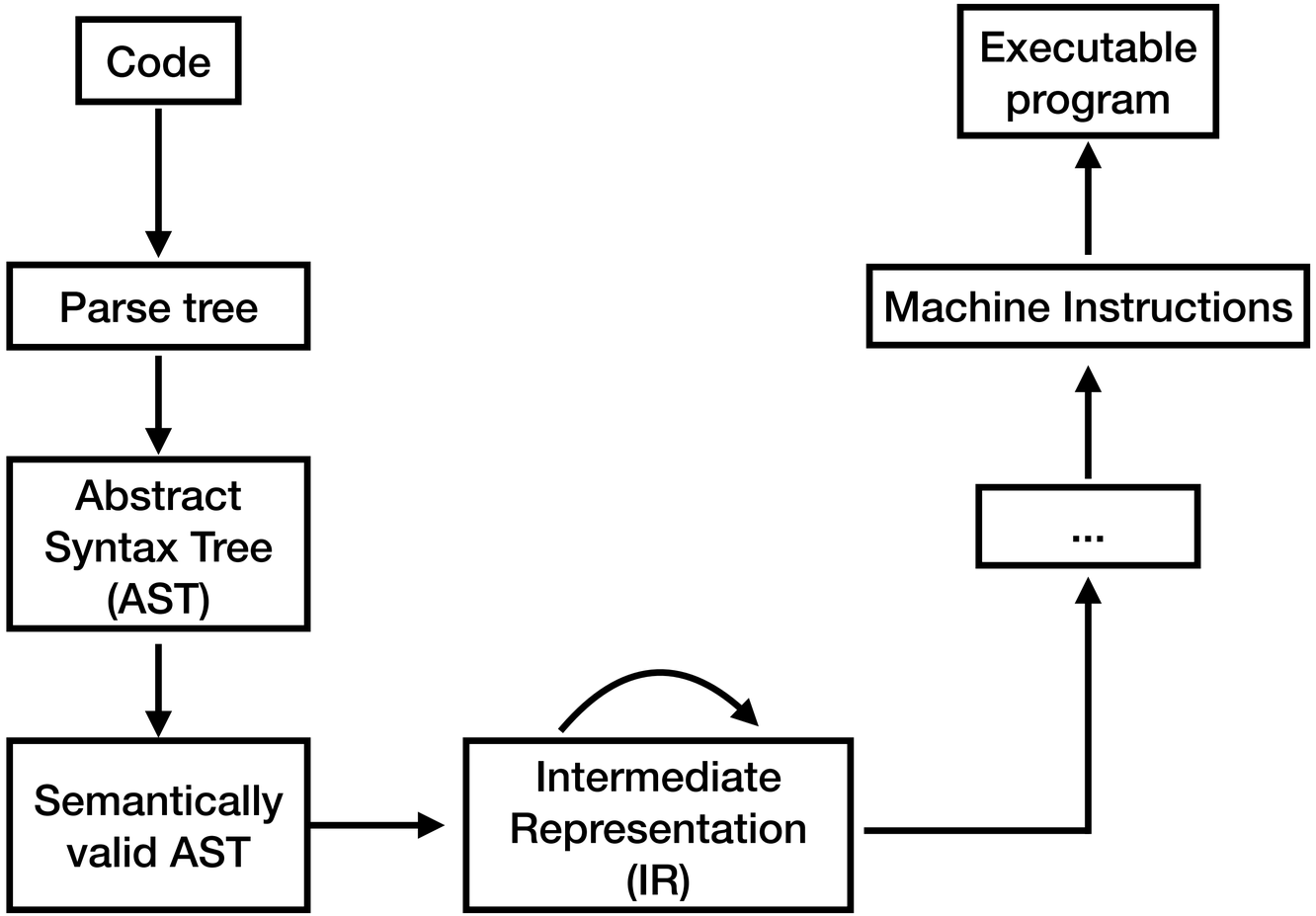

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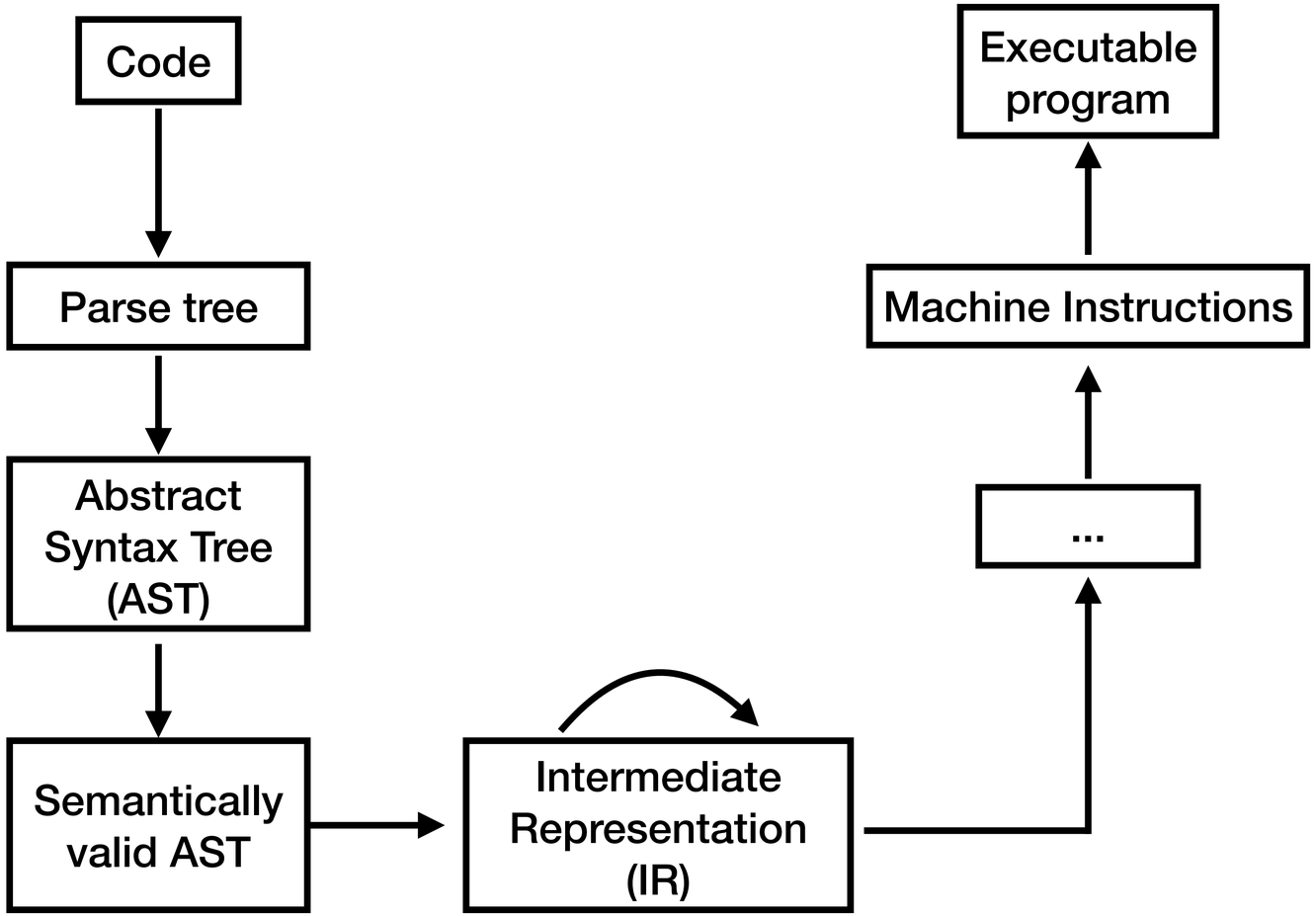

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